Economics and Human Biology 51 (2023) 101310

Contents lists available at ScienceDirect

Economics and Human Biology

journal homepage: www.elsevier.com/locate/ehb

Air quality and employee performance in teams: Evidence from the NFL

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ARTICLE INFO

JEL classification:
I19
J24
J28
Z22

Keywords:
Air quality
Team productivity
Employee health

ABSTRACT

Emerging research documents the impact of poor air quality on employee performance in a number of settings, including sport. Unlike other settings, sports teams have limited ability to influence previously scheduled games, making air quality exogenous to unobservable factors affecting game outcomes. We link play-level data from National Football League (NFL) games to data from air quality monitoring stations located near NFL stadiums. Results using data from about 90,000 offensive plays in 900 NFL games indicate that yards gained per offensive play increase with $PM_{2.5}$ concentration. Defensive players are relatively more affected by air quality than offensive players.

1. Introduction

The 1990 Amendments of the Clean Air Act identified curbing threats to the environment and human health from air pollution and toxic air emissions as major policy priorities.[1] These priorities reflect both established and emerging research evaluating the effect of exposure to air pollution on health and economic outcomes. In terms of health, a substantial literature grounded primarily in epidemiology and economics finds an adverse effect of air pollution. Long-term exposure to common air pollutants like fine particulate matter 2.5 micrograms (μg) or less in diameter ($PM_{2.5}$) and ozone (O_3) generate a number of adverse health effects like increased morbidity from chronic conditions (for example, respiratory infections and asthma), premature mortality, and infant morbidity and mortality. Short-term exposure to air pollution is also associated with adverse physiological and mental health effects including compromised lung functioning, irregular heartbeat, nonfatal heart attacks and impaired cognitive ability.

The negative health impacts of exposure to air pollution translate into economic costs associated with health care expenditures, lost productivity due to sick days, and premature deaths. Perhaps in response to the well-documented adverse population health outcomes attributable to air pollution, a related emerging strand of literature seeks to understand the broader effects of air pollution exposure on society. Much of this research focuses on analyzing the effect of exposure to a range of air pollutants on productivity in physical jobs in agricultural and warehouse settings, and increasingly on cognitive jobs and team settings. Our paper adds to this literature by analyzing the effect of exposure to air pollution on team productivity in the context of the National Football League (NFL).

We contribute to the literature analyzing team production using detailed, play level data from sport (Bryson and Papps, 2016; Kuehn, 2017; Arcidiacono et al., 2017; Schmidt, 2021; Wegelin et al., 2022) by analyzing the impact of randomly assigned negative productivity shocks on the production of team members exposed to this shock. The results show that reduced workplace air quality affected the production of different teams, and presumably individual members of these teams, differently, emphasizing the importance of heterogeneity in team goals and roles in determining the impact of negative productivity shocks on team production.

We also contribute to the literature analyzing the impact of poor air quality on employee performance by extending the analysis to detailed, team level outcomes, an important extension since a majority of jobs now require employees to work in teams. Previous research analyzed individual athlete or referee performance in a team sport setting (Lichter et al., 2017; Archsmith et al., 2018) or the performance of individual athletes (Guo and Fu, 2019; Mullins, 2018; Granella, 2021; Naidenova et al., 2022).

The results, based on analysis of about 90,000 offensive plays in roughly 900 NFL games played over the 2008 through 2016 seasons, indicate that yards gained per offensive play, and in particular yards gained on passing plays, increase with $PM_{2.5}$ concentration, suggesting that defensive team production is relatively more sensitive to bad air quality than offensive team production. One potential mechanism is that offensive players take on different roles than defensive players

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[1] https://www.epa.gov/clean-air-act-overview/clean-air-act-highlights-1990-amendments, accessed 06/30/2020.

https://doi.org/10.1016/j.ehb.2023.101310
Received 29 December 2022; Received in revised form 22 September 2023; Accepted 2 October 2023
Available online 5 October 2023
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in team production. Offensive players have predetermined assignments to perform on each play while defensive players mostly react to the planned plays of the offense. It is plausible that poor air quality has a greater negative impact on how defensive players react, and on their reaction time.

Results show an increase of about 4.3 additional passing yards over the course of a game for a one standard deviation increase in $PM_{2.5}$ particulate matter in the stadium. This represents about a 1.8% increase in passing yards. The impact occurs primarily in outdoor football stadiums. The literature on air quality and athletic performance generally contains small estimated effect sizes and our small, but precisely estimated results likely reflect the fact that air quality on game day seldom falls in the unhealthy range. Teams of offensive and defensive football players pursue different goals and fill different roles in team production. The results highlight the importance of team heterogeneity in terms of goals and individual player roles.

2. Context

A growing body of literature analyzes the short run impact of air quality on health, worker productivity, and other economic outcomes. The mechanisms through which air quality affects employee performance include physiological effects, cognitive effects, and employee behavioral responses to poor air quality. Physiological impacts include reduced cardiovascular and pulmonary functioning. Cognitive impacts have a physiological source – brain stress, cerebro-vascular disfunction, and impacts on the blood–brain barrier in the central nervous system – that manifest as mental fatigue, concentration loss, and impaired judgement (Archsmith et al., 2018). Employee behavioral responses to poor air quality include reduced effort, reduced labor supply (calling in sick), absenteeism, and reduced labor force participation.

A number of pollutants contribute to reduced air quality, including fine particulate matter of diameter less than 2.5 μg ($PM_{2.5}$), particulate matter of diameter 10 μg or less (PM_{10}), sulfur dioxide (SO_2), nitrogen dioxide (NO_2), carbon monoxide (CO), and ozone (0_3). Long-run exposure to elevated levels of these pollutants lead to negative health outcomes like cardiopulmonary disease, respiratory infection, lung cancer, asthma, and reductions in life expectancy (Graff Zivin and Neidell, 2013). A related strand of this literature shows that pollutants affect short-run economic outcomes like worker productivity and labor supply in settings as diverse as outdoor agricultural workers (Graff Zivin and Neidell, 2012), indoor factory workers (Chang et al., 2016), debating politicians (Heyes et al., 2019), and call center workers (Chang et al., 2019).

Several papers examining the short-run impact of poor air quality on worker productivity use data from sports. Archsmith et al. (2018) analyzed the impact of carbon dioxide and $PM_{2.5}$ concentration on the performance of MLB umpires. While umpiring MLB games requires relatively little physical activity, it requires substantial cognitive activity. Increased concentrations of both pollutants impacted umpire performance, increasing the likelihood that an individual game participant, the home plate umpire, incorrectly called a ball or strike by as much as 11%. The impact of $PM_{2.5}$ concentration appeared approximately linear while the impact of carbon monoxide increased substantially at very high concentrations.

In a closely related paper, Lichter et al. (2017) analyzed the impact of PM_{10} airborne particulate matter and ozone on the performance of individual football players in a top German football league, the Bundesliga, over the 1999 through 2008 seasons. Football is a team sport, but Lichter et al. (2017) analyzed individual player outcomes in terms of passes per individual player over the course of a match. While increased ozone concentration generated no impact on productivity, a one standard deviation increase in PM_{10} concentration generated a reduction in 2.4% of a standard deviation in passes per match, a small effect. Lichter et al. (2017) reported larger effects at very high PM_{10} concentrations, suggesting a non-linear effect of particulate matter on employee productivity. Guo and Fu (2019) pointed out that analyzing individual-level production in a team sport could overstate the impact of air quality on employee productivity if a free-rider problem exists where a football player tiring because of poor air quality reduces effort and instead relies on the effort of teammates.

Guo and Fu (2019) analyzed the impact of air quality, as reflected in an index composed of $PM_{2.5}$, PM_{10}, SO_2, NO_2, CO, and O_3 concentrations, on runners' marathon times in a large number of foot races in China in 2014 and 2015. The air quality index reflects the maximum concentration of the six pollutants on each day. The results indicate that a 10% increase in the air quality index increased the average finishing time by 1.1 min. Along similar lines, Mullins (2018) analyzed the impact of ozone on the performance of individual athletes in intercollegiate track and field competitions and found reduced performance in athletes competing in endurance events when ozone concentration increased.

Klingen and van Ommeren (2020) analyzed the impact of ozone on cycling speeds in London using data on more than 40 million trips made on rental bicycles over the years 2013–2017. The data contain a time stamp and location for the start and end of each bicycle trip, and the authors estimated distance traveled as the shortest road route between trip start and end locations. Note that this paper did not directly analyze cycling speed because they lacked information on stops made along each trip. The results showed that a 10 parts per billion increase in ozone concentration lead to a 0.3–0.4% reduction in cycling speed, a reduction of 53 m traveled per hour at the sample average speed. While informative, note that the data analyzed reflect a recreational or commuting activity rather than a competitive sport.

Naidenova et al. (2022) analyzed the impact of ozone and $PM_{2.5}$ particulate matter on the performance of participants in Ironman Triathlon races, which consists of long distance swimming, cycling, and running, over the period 2005 to 2019. This setting allows for estimation of heterogenous effects of air pollution on performance. The paper finds that high ozone levels increase swimming times and high $PM_{2.5}$ increases cycling and running times.

This literature analyzes the impact of air quality on the performance of individual athletes. No literature investigates the impact of air quality on team outcomes. Since most workers operate in a team environment, evidence on the impact of air quality on team performance will extend research in this area in an important way.

3. Empirical analysis

3.1. Measuring NFL productivity

An analysis of the relationship between local air quality and worker productivity requires a quantitative measure of production. Previous research developed several quantitative measures of individual worker production or performance. Chang et al. (2016) analyzed data on pear packers at a single processing plant compensated on a piece-work basis using data on observed pears packed per shift worked, a worker production variable. In the sports context, Lichter et al. (2017) proxy for individual worker production using the number of passes made by individual football players per game in the Bundesliga football league in Germany. Guo and Fu (2019) analyzed individual production, the finishing time for millions of runners competing in marathon races in China. Mullins (2018) analyzed outcomes in a number of different intercollegiate track and field competitions.

The NFL represents a team sport characterized by highly interrelated performance of eleven specialized workers who all contribute to team production. While some individual-level performance measures exist in the NFL, for example the total number of rushing yards gained by individual running backs in a game or the number of passing yards completed by a quarterback in a game, these totals reflect team effort, and not just individual production. For example, rushing yards gained by a running back depends heavily on the performance of the offensive

linemen. Because of this, the baseline measure of productivity used here is observed at the team level: the number of yards gained or lost on an individual offensive play from scrimmage.

The goal of football teams, winning games, means scoring more points than the opposing team. Yards gained per offensive play is affected by several interrelated factors including the physical and mental effort by the eleven offensive players and eleven defensive players on the field, play call, play development and execution, and other random factors. As yards gained per play increases, the offensive team effort or production exceeds that of the defense. As yards gained per play decreases, the opposite occurs. So yards gained or lost per play reflects production by the offensive team on the field relative to production of the defensive team on the field at each point in the game. This measure of production (yards gained or lost per play) does not assume that the set of moves that comprise a play are independent but the outcome of a play is plausibly independent from the outcome of another play.

Measuring production at the individual play level provides other empirical advantages. A play is the smallest discrete unit of action in a football game. It has a clear beginning and end. A fixed number of players participate in each play, since NFL teams cannot substitute within a play. A number of observable and sometimes related characteristics that plausibly affect effort or production on an individual play exist, including the down, yards needed to get a first down, the current game score, the position on the field, the amount of time left in the game, and the number of plays elapsed since the start of the current drive. These observable factors vary across plays and can be used to explain observed play outcomes holding other factors constant. Other related papers also control for observable factors that could plausibly affect outcomes. As examples, Archsmith et al. (2018) estimate models with extensive pitch, game, and umpire level control variables and Lichter et al. (2017) estimate models with large numbers of match and player-specific control variables.

Our empirical approach fits in the literature on estimating team production functions using play-level data from sports. Much of this literature focuses on motivating and explaining the presence of within-team productivity spillovers in production. In this context, the use of play-level data permits the isolation of individual effects on team performance based on player substitutions. Bryson and Papps (2016) develop a theoretical model of team production in sport and estimate structural parameters related to output spillovers using play-by-play level data from Major League Baseball games.

Arcidiacono et al. (2017) estimate an empirical model of team outcomes in the National Basketball Association (NBA) that accounts for the individual players on the court for individual plays in games. They exploit this variation to estimate the impact of spillovers on observable play-level team outcomes measured as the probability that a specific player scores on a team possession conditional on the other players on the court at that time and their abilities. Arcidiacono et al. (2017) point out that scoring on an individual play may not contribute to the team objective, winning games, and develop a measure of expected possession outcomes to account for this. Kuehn (2017) extends this research using team play-by-play outcomes in the NBA.

The papers in this literature analyze team-level outcomes at the individual play level and focus on developing evidence that spillovers among individual team members impact team-level outcomes. We also analyze team-level outcomes at the individual play level. We abstract away from the contribution of individual players and instead focus on the impact of a randomly assigned environmental factor, air quality, that represents a negative productivity shock potentially affecting the performance of all team members in each play observed in the data.

We implicitly assume heterogeneity in players, their performance, and their roles in producing team outcomes, in terms of the impact of exogenous variation in air quality across games on player performance. Also, offensive and defensive teams perform different tasks and have different objectives. Air quality may affect these teams differently, generating heterogenous impacts. This allows us to interpret the outcome of each individual play as reflecting the relative impact of air quality on the offensive and defensive players participating in each play. Since air quality is assigned to all players equally, random variation in air quality in football stadiums would have no effect on team play level outcomes.

Some of the measured effect of air quality on play outcomes could also be attributed to unobservable behavioral responses to the effects of pollution by the opposing team. Suppose a player is adversely affected by pollution and makes an obvious mistake on a play. The opposing team may observe this and change its behavior in an attempt take advantage of additional mistakes by this player. In this case, the measured effect of poor air quality reflects not only the physiological effect of pollution but also any behavioral responses to those effects.

3.2. Empirical approach and identification

We establish that air quality affects play outcomes in American football games. We analyze data at the most granular level available: the outcome of individual plays. Plays are discrete events beginning with the snap of the ball and ending when the player with the ball is ruled down, goes out of bounds, crosses the goal line, or attempts a forward pass that falls incomplete. Each play p results in a number of yards (Y_p) gained ($Y_p > 0$) or lost ($Y_p < 0$). The number of yards gained or lost on a play depends on effort and actions taken by the offense (the team possessing the ball) and defense. We estimate a reduced form linear model explaining observed variation in yards gained or lost on each play in NFL game g played involving offensive team o and defensive team d on date t in week w of season s:

$$Y_{pgodtws} = \gamma_o S_{so} + \gamma_d S_{sd} + \gamma_w + \gamma_s + \gamma_q + \gamma_p X_{pgod} + \beta PM_{gtws} + \varepsilon_{pgodtws}. \quad (1)$$

The explanatory variable of interest, PM_{gtws}, reflects the concentration of some airborne particulate matter at game g played on date t in week w of NFL season s. NFL teams do not control the schedule, which is also set months in advance, and cannot decide to play or not play a game depending on the air quality on game day. These factors should make the air quality on game day unrelated to unobservable factors affecting the number of yards gained or lost on plays. This forms the basis of our identification strategy which is similar to other papers in the literature.

The parameter of interest, β, reflects the impact of game day air quality on yards gained or lost on each play in game g. A negative estimate of β means bad air quality affects the offensive team more than the defensive team, resulting in fewer yards gained (or more yards lost) on each play, holding other factors constant. A positive estimate of β means bad air quality affects the defensive team more than the offensive team, resulting in more yards gained (or fewer yards lost) on each play, holding other factors constant. A zero estimate of β means bad air quality affects both the defensive and offensive teams equally, or has no impact on play outcomes.

S_{so} and S_{sd} contain full offensive team by season and defensive team by season interaction terms that capture unobserved offensive team and defensive team characteristics constant over the course of season s that affect yards gained (or lost) on each play like the quality of players and coaches. γ_w is a vector of week-of-season indicator variables that capture unobservable factors affecting yards gained (or lost) on each play associated with the progression of the season like general player fatigue and injuries and pressure during the end of the season playoff race. γ_s is a vector of season indicator variables. γ_q is a vector of indicator variables for the quarter of play in each game that capture within-game unobservable factors affecting yards gained (or lost) like within-game player fatigue.

X_{pgod} is a vector of play specific factors that might affect yards gained or lost on each play and vary across plays. These variables include the number of additional yards needed for the offensive team

to get a first down, the number of yards to the goal line, the current score difference between the offensive team and the defensive team (expressed as a positive number when the offensive team has the lead), an indicator variable for games played in enclosed stadiums, and a variable reflecting the current play number on each drive. This variable equals 1 on the first play of a drive, two on the second play of each drive, and so on. $\varepsilon_{pgodtws}$ is an error term that captures all other factors that affect yards gained or lost. We assume a heteroscedastic, mean zero $\varepsilon_{pgodtws}$ that exhibits within-game correlation, which requires cluster correction at the game level.

3.3. Data

3.3.1. Air quality data

Data on local air quality come from the US Environmental Protection Agency's (EPA) Air Quality System (AQS) database. The EPA operates thousands of ambient air quality monitoring stations throughout the county. The AQS database contains the exact location of each AQS monitoring station, in terms of latitude and longitude. This location information can be used to calculate the distance between each NFL stadium in the sample and the nearest AQS monitoring station. Following Archsmith et al. (2018), we use air quality data from the closest monitoring station within 10 miles of each stadium. 22 of the 31 NFL stadiums in use over the 2008–2016 period had an AQS monitoring station within 10 miles. The EPA changes the location of AQS monitoring stations over time, so we do not have air quality data for each game played at these 22 stadiums. Table 10 in the Appendix shows the distribution of games with air quality data across teams and stadiums in the sample.

These monitoring stations collect data on the concentration of a number of pollutants at high frequency, typically hourly measurements. We focus on two pollutants known to affect human mental and physical functioning in existing research: $PM_{2.5}$ particulate matter and ozone (O_3). $PM_{2.5}$ particulate matter consists of fine droplets or particles less than or equal to 2.5 μm in the air. An inch contains 25,000 μm. This equates to a size about thirty times smaller than a human hair. The EPA AQS data base measures $PM_{2.5}$ particulate matter concentration in micrograms per cubic meter of air (μg/m^3).

We focus on these pollutants for two reasons. First, evidence suggests that short term exposure to $PM_{2.5}$ affects both mental acuity and physical health. Second, Cusick et al. (2023) find that $PM_{2.5}$ and O_3 drive about 91 percent of the EPA's calculated Air Quality Index (AQI) which is based on five pollutants (ground level ozone, particulate pollution, carbon monoxide, sulfur dioxide, and nitrogen oxide) (AirNow.gov, 2023). $PM_{2.5}$ particulate matter can penetrate indoor spaces through a building's air circulation system.

These two pollutants represent the most common forms of air pollution analyzed in the literature on air quality and employee performance. Previous research links elevated levels of $PM_{2.5}$ particulate matter to reduced employee performance in many settings, both indoors and outdoors. Chang et al. (2016) find that performance of workers in a pear packing plant deteriorated during periods of high $PM_{2.5}$ particulate matter concentration. Archsmith et al. (2018), Lichter et al. (2017) and Guo and Fu (2019) find evidence that high concentrations of $PM_{2.5}$ particulate matter reduced the performance of athletes and referees in sporting events.

We observe daily $PM_{2.5}$ particulate matter concentration on an hourly basis for 22 NFL stadiums over the 2008 to 2016 seasons. However, our NFL play-by-play data do not include the start time of games, so we cannot easily calculate the average $PM_{2.5}$ particulate matter concentration while the games are being played. Most NFL games take place on Sunday and start at either 1 pm or 4 pm eastern time. Teams play a single game on Sunday night at 8 pm eastern time, a single game at 8 pm eastern time on Monday night, and a single game at 8 pm eastern time on Thursday night. The NFL schedule also includes games on Thanksgiving Day, a Thursday. Since we lack access to start time, we use the average $PM_{2.5}$ particulate matter concentration over the period 12 pm to 6 pm local time as our measure of air quality. This period includes almost all games played on the NFL schedule.

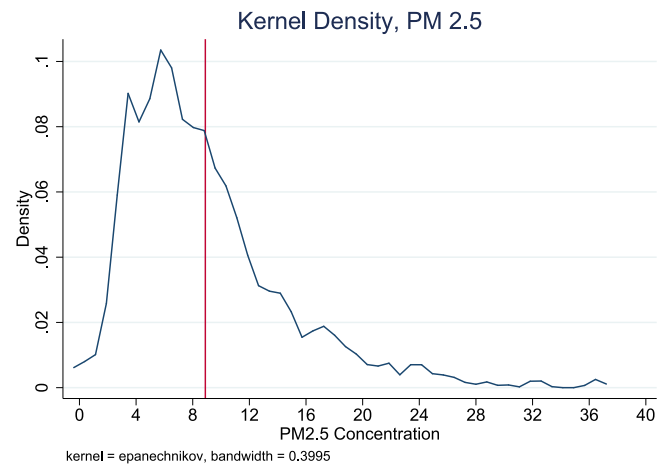

Fig. 1. Game day distribution of PM 2.5. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

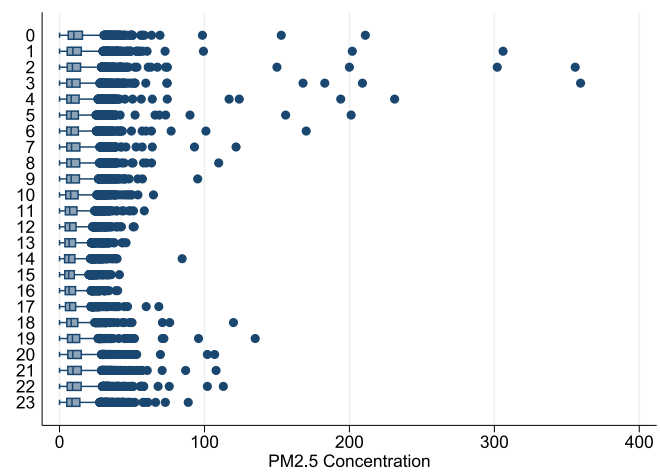

Fig. 2. Game day PM 2.5 concentration by time of day.

Fig. 1 shows the estimated kernel density of the distribution of the $PM_{2.5}$ particulate matter concentration variable on game days. The variable has a relatively long right tail which generates some skew in the distribution. The red line is drawn at the mean of the distribution, a concentration of 8.89 μg/m^3.

Variation in $PM_{2.5}$ particulate matter is the explanatory variable of primary interest in the regression models. Since we average $PM_{2.5}$ concentration over a period of hours, we first document the distribution of hourly $PM_{2.5}$ concentration on game days. Fig. 2 contains box-and-whisker plots for hourly $PM_{2.5}$ concentration from midnight (0) to 11 pm (23) in the sample. The box represents the 25th to 75th percentiles of the distribution and the line through the box is the median. The whiskers define the Tukey Fence, which are the 25th and 75th percentile values plus and minus 1.5 times the interquartile range. Dots identify extreme values outside the Tukey Fence.

From Fig. 2, the small boxes indicate relatively tight variation in the interquartile range. The left whisker is bounded at zero. The distribution shows a number of extreme values in early morning (midnight-6 am) and late evening (6–11 pm) hours, but not during daylight hours. Most of the extreme values appear concentrated in the midnight-6 am period, far outside the time window for NFL games. Averaging over the noon–6 pm period is not affected by extreme values.

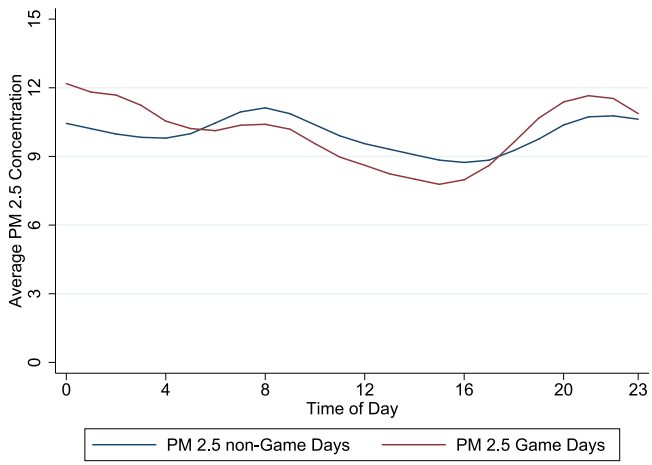

Fig. 3. Game day and non- game day PM 2.5 concentration.

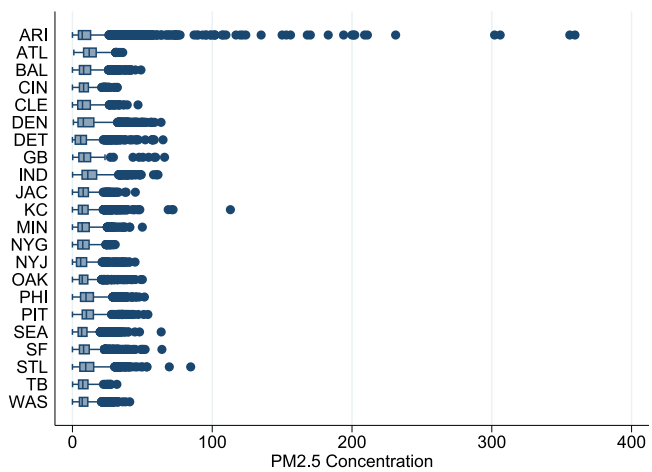

Fig. 4. Game day PM 2.5 concentration by team.

Fig. 3 shows $PM_{2.5}$ concentration by time of day on game days and non-game days in the sample. Daytime $PM_{2.5}$ concentration on game days is slightly lower than on non-game days. While a formal t-test of the difference rejects the null of equal means of game day and non game day $PM_{2.5}$ concentration, the average difference is about 0.1 μg/m^3, or about 1/100th of the mean.

Fig. 4 shows the distribution of hourly $PM_{2.5}$ particulate matter concentration by home team, which is equivalent to the distribution by stadium except for the New York Jets and Giants, who share a stadium. Games played in the University of Phoenix Stadium, home of the Arizona Cardinals (ARI on Fig. 4) feature many more extreme values than games played in other stadiums, likely reflecting the hot dry climate in Arizona. The distribution of $PM_{2.5}$ particulate matter concentration at other stadiums appear to be relatively similar.

Wind speed can affect $PM_{2.5}$ concentration. Hart et al. (2020) find that low wind speed is associated with higher concentration of $PM_{2.5}$. On the other hand, extreme wind represented by wind gusts reduces $PM_{2.5}$ concentration due to more atmospheric turbulence that promotes diffusion and dilution of pollutants. We can control for the effect of wind speed on $PM_{2.5}$ concentration because the play-by-play data contains information on temperature, wind speed, and wind direction for games played in open-air stadiums. It is not clear how to account for the impact of wind speed on $PM_{2.5}$ concentration in domed and closed roof stadiums, so we limit our analysis of wind speed effects on pollutant concentration to a subset of open air stadiums.

Table 1
Summary statistics, play-level data.

	PASS	RUSH	Total
PM2.5 concentration	8.89	8.92	8.90
	(5.65)	(5.59)	(5.63)
Yards gained	6.45	4.20	5.49
	(10.23)	(6.42)	(8.89)
Score difference	−2.93	0.83	−1.33
	(10.40)	(10.64)	(10.67)
Yards to go	8.92	8.28	8.65
	(3.94)	(3.82)	(3.90)
Yards to goal line	53.47	52.18	52.92
	(23.91)	(25.50)	(24.61)
Quarter	2.58	2.54	2.57
	(1.12)	(1.15)	(1.13)
Consecutive plays	4.33	3.84	4.12
	(2.87)	(2.85)	(2.87)
Observations	65,421	48,361	113,782

Notes: Means and standard deviation (in parentheses) are reported for variables used in baseline specification.

3.3.2. NFL play-by-play data

Data on the outcomes of individual plays in NFL games come from ArmchairAnalysis.com, a sports data provider.[2] The data file contains detailed information on every play from every NFL game played beginning in the 2000 season. Each record in the data set reflects a play of some type and identifies game characteristics at the start of each play.

The empirical analysis focuses on only offensive rushing or passing plays that did not result in a turnover. We ignore turnovers because the defensive team gains yards on these plays. Incomplete passes are coded as gaining zero yards. Penalty yards added to the end of a play are excluded. Kickoffs, punts, field goal attempts, plays blown dead because of penalties, kneel-downs, and extra point attempts are not included in the sample. Excluding these plays generates relatively homogenous group of plays for analysis.

Table 1 summarizes the play-level data analyzed. The sample contains 113,782 plays, including 65,421 plays in which a pass was attempted and 48,361 rushing plays. The $PM_{2.5}$ concentration is similar on passing plays (8.89) and rushing plays (8.92). The average passing play gained about 6.5 yards (including incomplete passes) and the average rushing play gained 4.2 yards. The score difference at the start of the play reflects points by the offensive team minus points by the defensive team. The average passing play occurred when the offensive team trailed and the average rushing play when the offensive team had a small lead. Trailing teams pass more and teams with a lead run more.

The average passing and rushing play occurred with about 8 yards to go to get a first down. The ball was on the team's own 47 yard line on the average play. More plays occurred in the second half than in the first half.

Yards per play represents the key outcome variable in our analysis. Fig. 5 shows kernel density estimates for the yards per play variable for the entire sample and split by passing plays and running plays. The red lines represent means. All have long right tails and some skew, representing a small number of plays that gain significant yardage. The median rushing play gained 3 yards with an interquartile range of 5 yards (1 to 6 yards). The median passing play gained 4 yards with an interquartile range of 10 yards (0 yards to 10). The outcome of passing plays is more variable than the outcome of rushing plays.

The passing yards variable is bimodal, with peaks at zero and at the sample average. This reflects the presence of a large number of incomplete passes in the data, which gain zero yards.

[2] ArmChairAnalysis.com was acquired by FTN Data in 2020.

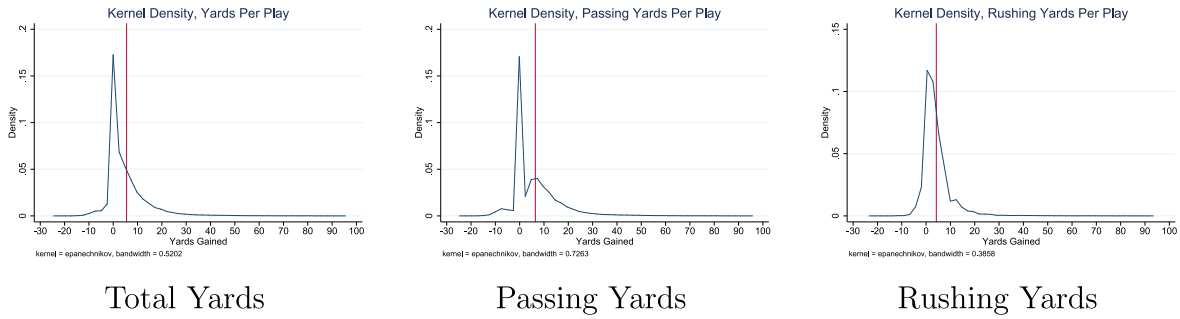

Fig. 5. Kernel densities, Yards per play. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Summary statistics - Game level.

Variable	Mean	Std. Dev.	Min	Max
Pass yards per play	6.45	1.26	3.07	10.85
Run yards per play	4.20	0.92	1.39	8.05
Plays per game	126	8.85	104	167
Rush plays per game	54	7.20	31	79
Pass plays per game	73	10.87	42	113
Total passing yards	465	116.8	187	826
Observations	908			

Table 3
Regression results - PM 2.5 and yards per offensive play.

	All	Rush	Pass
PM2.5 concentration	0.014**	0.00064	0.021*
	(2.69)	(0.12)	(2.50)
Score difference	−0.039***	−0.031***	−0.0073
	(−12.4)	(−9.19)	(−1.57)
Yards to go	0.11***	0.11***	0.063***
	(15.1)	(12.6)	(5.72)
Yards to goal line	0.036***	0.023***	0.044***
	(26.3)	(15.3)	(19.7)
Quarter	−0.051*	−0.11***	0.011
	(−2.23)	(−4.11)	(0.31)
Consecutive plays	0.020	0.042***	−0.011
	(1.73)	(3.36)	(−0.61)
Observations	113 737	48 316	65 421
R^2	0.021	0.030	0.025

Robust t-statistics in parentheses. *: 5%, **: 1%, ***:<1%.
Dependent variable: yards per offensive play. All models include the following fixed effects: offensive team by season, defensive team by season, week of season, season. A dummy variable for games played in enclosed stadiums is included.
Standard errors are cluster corrected at the game level.

Table 2 contains summary statistics calculated at the game level to provide context for the play-level data. The analysis sample contains about 900 games. The average game in the sample contained 126 plays, including 73 passing plays and 54 passing plays. The average game in the sample featured about 465 yards of passing.

3.4. Results

Eq. (1) represents the baseline empirical model, a linear reduced form model explaining variation in yards gained on offensive plays in NFL games. All estimated standard errors are robust to heteroscedasticity and cluster corrected at the game level to account for within-game correlation in the equation error term. Again, the parameter of interest is β reflecting the effect of local $PM_{2.5}$ concentration on yards gained or lost on a play during the game. Recall all models contain offensive team by season and defensive team by season fixed effects, week of season fixed effects, season fixed effects, and a dummy variable for games played in enclosed stadiums.

Table 3 contains baseline results. Results in the first column reflect the full sample of plays. Note the parameter estimates on the game condition variables are almost all statistically different from zero and appear reasonable in terms of estimated signs. The sign on the current score variable is negative, implying that plays run by teams trailing on the scoreboard gain fewer yards. The estimated parameter on the yards to go variable is positive, implying that the more yards needed to get a first down, the larger the gain on the play. The farther the spot of the ball from the goal line, the more yards gained on the play. This reflects the fact that as a team gets closer to the goal line, it simply has less ground over which to gain yards. Yards gained decline with the quarter of the game, likely reflecting that fatigue affects the offense more than the defense in NFL games. The longer a drive lasts, the larger the gain on each additional play run by the offense, likely indicating increasing fatigue among the defensive players.

The parameter estimate on the $PM_{2.5}$ concentration variable is positive and statistically different from zero. This sign implies that the effect of airborne fine particulate matter affects the defense more than the offense in NFL games. Teams of offensive players undertake different actions and have different objectives than teams of defensive players. Offensive plays represent scripted action on the part of the team of offensive players. Each offensive player has a detailed, specific assignment to carry out on each play so he knows where he is supposed to run and what he is supposed to do. On the other side of the line of scrimmage, defensive players primarily react to the planned actions of the offensive players. Defensive players do not necessarily know where they will need to run, or what actions they will need to take, on any play. Thus air quality could affect these players differently.

The dependent variable is measured in yards, so the size of the parameter estimate on the $PM_{2.5}$ concentration variable, 0.014, implies that for each additional microgram of $PM_{2.5}$ particulate matter per cubic meter of air near the stadium, the offensive team gains about a 0.5 inch of additional yardage on a given play. The average play gained 5.49 yards, or 198 inches, so this represents a 0.2% increase in yards gained.

From Table 2 the average game featured 126 plays. From Table 1, the standard deviation of airborne $PM_{2.5}$ particulate matter was 5.63. At the sample average, a one standard deviation increase in airborne $PM_{2.5}$ particulate matter generated an additional 5 yards of offense over the course of a game for each team, assuming that each team ran half of the 126 plays.

The second and third columns on Table 3 estimate separate regression models for rushing and passing plays. Rushing and passing plays differ in fundamental ways, in terms of the distance they cover.

Table 4
Regression results - Nonlinear PM 2.5 and yards per offensive play.

	All	Rush	Pass
PM 2.5 15–27	0.097	−0.00074	0.12
	(1.16)	(−0.0085)	(0.81)
PM 2.5 27+	0.61*	0.26	0.72
	(2.02)	(0.87)	(1.53)
Score difference	−0.037***	−0.029***	−0.0089
	(−12.0)	(−8.86)	(−1.94)
Yards to go	0.093***	0.079***	0.074***
	(13.3)	(10.5)	(6.86)
Yards to goal line	0.039***	0.026***	0.042***
	(29.7)	(17.7)	(19.2)
Quarter	−0.048*	−0.096***	0.013
	(−2.11)	(−3.73)	(0.34)
Consecutive plays	0.065***	0.068***	−0.037*
	(5.97)	(5.70)	(−2.16)
Observations	113 737	48 316	65 421
R^2	0.021	0.029	0.025

Robust t-statistics in parentheses. ∗: 5%, ∗∗: 1%, ∗∗∗: <1%.
Dependent variable: yards per offensive play. All models include the following fixed effects: offensive team by season, defensive team by season, week of season, season. A dummy variable for games played in enclosed stadiums is included.
Standard errors are cluster corrected at the game level.

Different rules apply to rushing and passing plays. For example, an offensive lineman cannot run down field beyond the line of scrimmage on a passing play, but he can run down field on a running play. These differences mean that fine airborne particulate matter could affect NFL players differently on running and passing plays. The results on Table 3 suggest that this is the case.

From Table 3, higher concentrations of $PM_{2.5}$ particulate matter per cubic meter of air near the stadium increases only yards gained on passing plays and not on running plays. The parameter estimate of 0.21 represents about 0.75 inches, or about 4.3 additional passing yards over the course of a game for a one standard deviation increase in $PM_{2.5}$ particulate matter. This represents about a 1.8% increase in passing yards.

Although these effect sizes may seem small, they are in line with the literature on the impact of air quality on performance in sport which generally contains relatively small effect sizes. Archsmith et al. (2018) reported that an increase of one part per million (ppm) in carbon dioxide resulted in an 11% decrease in umpire correct calls, or two more incorrect calls per 100 calls made. But the sample average carbon dioxide concentration in that paper is 0.295 ppm with a standard deviation of 0.14, so a 1 ppm increase represents a 7 standard deviation increase in carbon dioxide concentration.

Lichter et al. (2017) found that a 1% increase in PM_{10} concentration led to a 0.02% decrease in average football passes per player per match, or a 2.4% of a standard deviation reduction for a 1 standard deviation increase in PM_{10} concentration. Guo and Fu (2019) reported that a 10% increase in the air quality index increased the average marathon finishing time (4 h 36 min) by 1.1 min.

The existing literature contains evidence of a non-linear impact of pollutants on employee performance and exertion (Guo and Fu, 2019; Klingen and van Ommeren, 2020). Nonlinearity can be addressed in several ways including binned values of the explanatory variable and log transformations. The results on Table 4 divide the observed game time $PM_{2.5}$ concentrations into three bins: 0 to 15 μg/m^3 (the omitted category), 15 to 27 μg/m^3, and above 27 μg/m^3. The EPA defines hazardous levels of $PM_{2.5}$ concentration above 35.4 μg/m^3. But only 300 plays in the sample from 1 game have this high a concentration. The 15–27 μg/m^3 contains about 11% of the plays in the sample and the over 27 μg/m^3 about 1% of the plays in the sample. Also, some government agencies use 27 μg/m^3 as a threshold for poor air quality.

The results on Table 4 provide some limited evidence of nonlinearity in impact. In the pooled model, average offensive yards per play increases only in games where $PM_{2.5}$ concentration exceeds 27 μg/m^3. The kernel density graphs above indicate that both $PM_{2.5}$ concentration and yards gained per play exhibit long right tails. Log transformations represent one method for addressing this issue. However, the presence of many zeros and negative values in the yards gained variable makes this transformation less than ideal, as it will drop many observations.

3.5. Robustness checks

3.5.1. Differences in game day and practicing conditions

Home and visiting teams may be differentially impacted by game day air quality due to differences in prior exposure. The intuition is that the visiting teams practice in their home cities during the week leading up to the game, generating a difference in exposure to particulate matter than the home team, practicing in the city where the game occurs, does not experience. Mullins (2018) finds evidence that this occurs using outcomes from foot races. This asymmetry in exposure to air quality could have a differential impact on home and visiting team performance.

We explore this possibility by estimating a variation of Eq. (1) that specifies different weather shocks for home team offense (defense) and away team offense (defense). We exploit variation in these air quality shocks to analyze the impact of game day air quality on the visiting and home teams separately as well as the impact of air quality when each team is on offense and defense.

We again estimate a reduced form linear model explaining observed variation in yards gained or lost on each play in NFL game g played involving offensive home team oh and defensive visiting team dv on date t in week w of season s:

$$Y_{pgodtws} = \gamma_o S_{so} + \gamma_d S_{sd} + \gamma_w + \gamma_s + \gamma_q + \gamma_p X_{pgod} + \beta_{oh} PMS_{oh,gtws} + \beta_{dv} PMS_{dv,gtws} + \varepsilon_{pgodtws}. \quad (2)$$

We estimate a similar model explaining observed variation in yards gained or lost on each play involving offensive visiting team ov and defensive home team dh:

$$Y_{pgodtws} = \gamma_o S_{so} + \gamma_d S_{sd} + \gamma_w + \gamma_s + \gamma_q + \gamma_p X_{pgod} + \beta_{ov} PMS_{ov,gtws} + \beta_{dh} PMS_{dh,gtws} + \varepsilon_{pgodtws}. \quad (3)$$

The key explanatory variables, $PMS_{oh,gtws}$ and $PMS_{dvh,gtws}$ reflect air quality shocks to home team, h, on offense oh and visiting team, v, on defense dv. The air quality shock variable for any team, home or visiting, is the difference between the average $PM_{2.5}$ concentration over the entire sample period (2008–2016) in each city home to an NFL team and the actual game day air quality in the city where the game is played. The average air quality shock was negative and similar in size for home and visiting teams: −1.25 for home teams with a standard deviation of 5.5 and −1.5 for visiting teams with a standard deviation of 5.6.[3]

Table 5 contains the results. The left panel shows results using plays when the visiting team was on defense and the home team was on offense. The home team air quality shock reduced home team performance, primarily on rushing plays, where the home team averaged about 1.85 fewer rushing yards per play. The visiting defense was also negatively affected, giving up more yards on rushing plays and all plays. The results in column (1) show that the defense was impacted more than the offense, supporting the claim made above. The signs and significance of all other covariates are basically unchanged from those reported on Table 3.

[3] Note that the sample size for the weather shock analysis is considerably smaller because we only have data on the air quality for cities where 22 NFL teams play, so we can only estimate the air quality the players are accustomed to for these 22 teams when they play on the road. In the other cities, there was no air quality monitoring station near enough to the stadium to provide a reliable estimate of actual air quality at the stadium.

Table 5
Regression results - PM 2.5 shocks and yards per offensive play.

	Visiting team on defense			Home team on defense		
	All	Rush	Pass	All	Rush	Pass
Home team PM 2.5 shock	−0.86	−1.85***	−0.37	0.91*	0.28	1.29
	(−1.94)	(−3.74)	(−0.62)	(2.39)	(1.10)	(1.92)
Visitor PM 2.5 shock	0.87*	1.85***	0.40	−0.87*	−0.27	−1.24
	(1.97)	(3.74)	(0.67)	(−2.32)	(−1.06)	(−1.85)
Score difference	−0.058***	−0.030***	−0.033***	−0.063***	−0.047***	−0.043***
	(−9.09)	(−4.13)	(−3.48)	(−9.70)	(−6.94)	(−4.20)
Yards to go	0.11***	0.10***	0.051*	0.11***	0.12***	0.064**
	(7.73)	(5.50)	(2.49)	(6.92)	(5.76)	(2.83)
Yards to goal line	0.037***	0.027***	0.045***	0.034***	0.024***	0.041***
	(12.9)	(8.07)	(9.40)	(11.2)	(7.16)	(8.99)
Quarter	−0.096	−0.14*	−0.022	−0.066	−0.013	−0.10
	(−1.85)	(−2.52)	(−0.27)	(−1.32)	(−0.24)	(−1.36)
Consecutive plays	0.0098	0.062*	−0.039	−0.015	0.041	−0.055
	(0.41)	(2.43)	(−0.97)	(−0.62)	(1.39)	(−1.52)
Observations	26 529	11 355	15 174	26 484	10 916	15 568
R^2	0.027	0.041	0.033	0.026	0.038	0.035

Robust t-statistics in parentheses. *: 5%, **: 1%, ***:<1%.
Dependent variable: yards per offensive play. All models include the following fixed effects: offensive team by season, defensive team by season, week of season, season. A dummy variable for gamess played in enclosed stadiums is included.
Standard errors are cluster corrected at the game level.

The right panel contains results using only plays when the home team is on defense and the visiting team is on offense. The parameters on the air quality shock variables are statistically different from zero in the model where the dependent variable is average yards per play on all plays. The significance indicates that both teams are impacted as expected by the air quality shocks. Home teams experiencing larger air quality shocks give up 0.91 more yards per play on defense and visiting teams gain 0.87 fewer yards on offense. An F-test for the null hypothesis that $-\beta_{ov} = \beta_{dh}$ in column (4), the home team defense is affected more than the visiting team offense, rejects the null at a p-value of 0.006. The air quality shock clearly affects the defense more than the offense in that model specification.

3.5.2. Controls for weather

Evidence suggests that $PM_{2.5}$ concentration can be affected by wind and temperature. The NFL play-by-play data contain wind and temperature data at kickoff for games played in outdoor stadiums. Wind and temperature can also affect average yards per play. Table 6 contains results from estimating Eq. (1) including game time wind speed and temperature for a sub-sample of games played in open air stadiums. Note that it is unclear how to handle temperature in domed stadiums, since the climate in these facilities is controlled through both air conditioning and heating, depending on outside conditions. Because of this, we do not estimate a pooled model with an indicator variable for games played outdoors.

The results on Table 6 confirm the full sample results on Table 3, in that higher concentrations of $PM_{2.5}$ particulate matter per cubic meter of air near open air stadiums increase only yards gained on passing plays and not on running plays. The parameter estimate on the $PM_{2.5}$ concentration variable on passing plays reported on Table 6 exceeds the parameter estimate reported on Table 3, suggesting a slightly larger effect in outdoor stadiums, controlling for game time wind speed and temperature.

Other weather conditions, for example rainfall, could affect yards per game and $PM_{2.5}$ concentration. Rainfall impacts $PM_{2.5}$ concentration less than other, larger airborne particulate matter (Zhang et al., 2018). We lack information on game day rainfall but have information on game time humidity. Humidity and rainfall are positively correlated. We estimate additional models for games in outdoor stadiums that include game time humidity. These models contain slightly smaller sample sizes due to missing humidity data for some games.[4] The parameter estimate on the $PM_{2.5}$ concentration variable remains unchanged when including the game day humidity variable. The parameter estimate on the humidity variable is negative and statistically different from zero in the passing yards model only. All results are robust to controlling for humidity.

Table 6
Regression results - Outdoor stadiums.

	All	Rush	Pass
PM2.5 concentration	0.015*	−0.0038	0.028*
	(2.15)	(−0.55)	(2.50)
Wind speed	−0.018*	−0.014	−0.021*
	(−2.43)	(−1.53)	(−2.02)
Temperature	0.013***	0.0013	0.023***
	(3.68)	(0.30)	(4.31)
Score difference	−0.045***	−0.034***	−0.021***
	(−11.8)	(−8.69)	(−3.59)
Yards to go	0.088***	0.083***	0.061***
	(10.6)	(9.41)	(4.68)
Yards to goal line	0.039***	0.026***	0.042***
	(24.1)	(15.2)	(15.2)
Consecutive plays	0.061***	0.068***	−0.040
	(4.70)	(4.85)	(−1.91)
Observations	77 383	33 365	44 018
R^2	0.022	0.033	0.028

Robust t-statistics in parentheses. *: 5%, **: 1%, ***:<1%.
Dependent variable: yards per offensive play. All models include the following fixed effects: offensive team by season, defensive team by season, week of season, season. Variables for wind speed and temperature are included.
Standard errors are cluster corrected at the game level.

3.5.3. Ozone concentration

Table 7 contains results for an additional robustness check, replacing the $PM_{2.5}$ concentration variable with the ozone (O_3) concentration. Existing research shows that higher ozone concentration primarily affects participants in endurance sports like marathon running (Guo and Fu, 2019) or long-distance intercollegiate track events (Mullins, 2018). Lichter et al. (2017) found no effect of ozone on passes made by

[4] Results are available on request.

Table 7
Regression results - Ozone and yards per offensive play.

	All	Rush	Pass
Ozone concentration	−1.32	−0.76	−0.84
	(−0.56)	(−0.34)	(−0.20)
Score difference	−0.060***	−0.027***	0.0060
	(−24.5)	(−12.6)	(1.50)
Yards to go	0.13***	0.081***	0.064***
	(23.6)	(15.4)	(6.53)
Yards to goal line	0.048***	0.024***	0.049***
	(43.0)	(24.5)	(22.6)
Quarter	−0.036	−0.11***	0.063
	(−1.89)	(−6.14)	(1.81)
Consecutive plays	0.15***	0.050***	−0.018
	(17.3)	(6.12)	(−1.08)
Observations	185 347	98 670	86 677
R^2	0.026	0.024	0.026

Robust t-statistics in parentheses. *: 5%, **: 1%, ***:<1%.
Dependent variable: yards per offensive play. All models include the following fixed effects: offensive team by season, defensive team by season, week of season, season. A dummy variable for games played in enclosed stadiums is included.
Standard errors are cluster corrected at the game level.

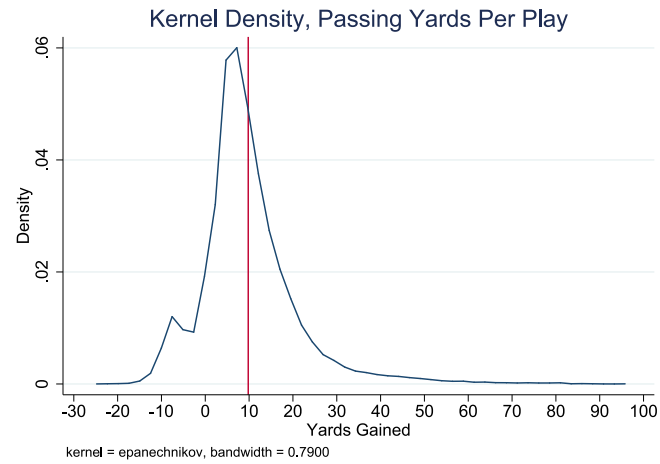

Fig. 6. Kernel density for passes with no incompletions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 8
Regression results - No incomplete passes.

	All	Pass	Incomplete pass LPM
PM2.5 concentration	0.020**	0.033**	0.0043
	(3.24)	(2.76)	(0.11)
Score difference	−0.071***	−0.0031	0.053**
	(−17.6)	(−0.50)	(2.81)
Yards to go	0.19***	0.048**	−0.34***
	(20.0)	(3.23)	(−7.02)
Yards to goal line	0.041***	0.053***	−0.10***
	(25.2)	(16.6)	(−9.24)
Quarter	−0.012	0.098	0.55**
	(−0.43)	(1.95)	(3.21)
Consecutive plays	0.057***	0.027	0.32***
	(4.13)	(1.07)	(3.39)
Observations	91 260	42 944	65 421
R^2	0.034	0.031	0.021

Robust t-statistics in parentheses. *: 5%, **: 1%, ***:<1%.
Dependent variable: yards per offensive play. All models include the following fixed effects: offensive team by season, defensive team by season, week of season, season. A dummy variable for games played in enclosed stadiums is included. Standard errors are cluster corrected at the game level.

German football players during matches. Also, ozone, a relatively large particle, may not affect football games played in indoor venues as much as $PM_{2.5}$, which can easily enter buildings (Chang et al., 2016).

The results on Table 7 confirm these findings in the literature. Variation in ozone concentration has no effect on offensive yards per play in the pooled and individual run play and pass play models. The parameter estimates on the play-level explanatory variables on Table 7 generally resemble those on the tables above, in terms of sign and statistical significance.

3.5.4. Setting incomplete passes to zero

Setting yards gained on incomplete passes to zero impacts the distribution of this variable. From Fig. 5, the yards per pass variable becomes bimodal in this case. This could affect the results in the paper. However, zero yards gained reflects the actual outcome of an incomplete pass. The key issue is the extent to which air quality affects incomplete pass rates.

Fig. 6 shows the distribution of the passing yards per play variable for only completed passes. Conditional on completion, the average passing play gains 9.8 yards, and the median passing play 8 yards. The interquartile range is 10 yards (4 to 14 yards) which is the same as that when incomplete passes are included.

Table 8 contains baseline regression results from Eq. (1) when incomplete passes are removed from the sample in Column 1 and 2. This again shows evidence that $PM_{2.5}$ concentration affects yard per play. The third column shows results for a linear probability model for a dependent variable equal to one red if a pass was incomplete and zero if complete, using the baseline explanatory variables. These results suggest that the impact of $PM_{2.5}$ concentration is not working through incomplete passes, since the parameter estimate on the $PM_{2.5}$ concentration variable is not statistically different from zero.

3.5.5. Additional robustness checks

Finally, we estimated models with alternative fixed effects and additional control variables. First, we estimated three versions of the baseline specification using an alternative set of fixed effects interacted at the city-week of season-season level, a substantially larger set of fixed effects than in the baseline model which controlled for city, week, and season fixed effects separately. We then add game/stadium level controls and then finally weather controls. The inclusion of these alternative fixed effects affects the parameter estimates on the weather variables, not $PM_{2.5}$. The parameter estimate on the $PM_{2.5}$ variable remains statistically different from zero and roughly the same size across all these alternative specifications.

We also estimated models with week-of-season-by-stadium fixed effects instead of week-of-season fixed effects. This had no qualitative effect on the parameter estimate on the $PM_{2.5}$ variable or the other game level control variables, but the estimated effect size of $PM_{2.5}$ is considerably larger than that reported on Table 3. Finally, we estimated a model that includes an interaction term between $PM_{2.5}$ and the indicator variable for a domed stadium. The parameter estimate on the $PM_{2.5}$ variable is larger, positive and significant. The parameter on the interaction term is negative (−0.023) and significant, so $PM_{2.5}$ particulate matter has a smaller effect in domed stadiums.[5]

4. Conclusions

Using detailed play level data from the National Football League, we estimate the effect of exposure to $PM_{2.5}$ and ozone O_3 on team productivity. The results, based on analysis of about 90,000 offensive plays in about 900 NFL games played over the 2008 through 2016

[5] Results from these robustness checks are available from the authors upon request.

seasons, indicate a non-negligible, positive effect on yards gained per offensive play, and in particular yards gained on passing plays, with increases with $PM_{2.5}$ concentration. Our causal identification relies on the exogeneity of air quality on game day since the NFL schedule is set months in advance. Given this, air quality on game day should be uncorrelated with unobservable factors affecting the number of yards gained or lost on plays.

The result that exposure to air pollution matters for yards gained, rather than yards lost suggests that air quality affects teams of workers with different roles and objectives differently. In our setting, defensive team production is relatively more sensitive to bad air quality than offensive team production. We posit that the mechanism underlying this result reflects that defensive players mostly react to the planned plays of the offense so poor air quality has a greater negative impact on defensive players' reactions.

We considered some options for identifying this mechanism including interacting the defense's overall experience with air quality and exploiting differences in probabilities of offensive and defensive penalties. We do not observe experience in our data so we cannot explore the interaction of experience with air quality idea. We do observe offensive and defensive penalties so we estimated linear probability models explaining the likelihood of an offensive (defensive) penalty on a play using the covariates in the yards per play models. Unfortunately, the parameter estimate on $PM_{2.5}$ is not statistically different from zero these models so we cannot identify our hypothesized mechanism using this approach.

A third alternative is to decompose yards gained on passing plays into yards before and after a catch; however, we could not explore this option because we only observe total yards gained on a completed pass. There is an alternative NFL data set at the play level that captures yards after a catch; however, there is not a common play identifier in both data sets that allows us to easily merge this variable with our analysis data. An avenue for future work is to use this alternative data set to conduct a similar analysis and attempt to identify our potential mechanism by decomposing pass yards into yards after catch and air yards.[6]

Our size effects are relatively small - a one standard deviation increase in $PM_{2.5}$ translates into approximately 4.3 additional passing yards per game. We do not find effects of ozone exposure on offensive yards per play. While these effects are small, they are consistent with findings in other papers analyzing the effect of air quality on player performance. The estimated effect sizes may be downward biased because we cannot directly control for the impact of air quality on coaching decisions. Archsmith et al. (2018) evaluate the effect of $PM_{2.5}$ concentration on the performance of home plate umpires in Major League Baseball and find that exposure to higher concentrations of $PM_{2.5}$ increased the likelihood of errors in calling balls and strikes. Unfortunately, we cannot test for similar effects with NFL coaches because we lack a observable measure of the optimality of a play call.

Our study adds to a growing literature studying the societal effects of air pollution exposure beyond health effects — in particular, effects on labor productivity and employee performance. We extend the literature by analyzing detailed team level outcomes in a setting with heterogenous teams. Our results from the NFL are useful for thinking about how exposure to ambient air pollution might affect firm outcomes in many settings where employees achieve results by working in teams. There are many such examples spanning a number diverse industries including the medical, information technology, academic, and financial sectors. Like our setting, team heterogeneity exists in these workplaces.

[6] We thank an anonymous referee for these suggestions and pointing to this alternative data set available through the "nflfastR" R package (or at GitHub: https://github.com/nflverse/nflverse-data/releases/tag/pbp).

Table 9
Sample coverage - Seasons and months of season.

Season	# games	% games	Month of season	# games	% games
2008	7	0.77	September	178	19.60
2009	46	5.07	October	213	23.46
2010	89	9.80	November	221	24.34
2011	95	10.46	December	237	26.10
2012	106	11.67	January	53	5.84
2013	125	13.77	February	6	0.66
2014	139	15.31			
2015	162	17.84			
2016	139	15.31			

Table 10
Sample coverage - Teams.

Team	Games visitor	%	Games home	%
ARI	37	4.07	57	6.28
ATL	23	2.53	47	5.18
BAL	28	3.08	71	7.82
BUF	24	2.64		
CAR	29	3.19		
CHI	30	3.30		
CIN	30	3.30	32	3.52
CLE	30	3.30	31	3.41
DAL	28	3.08		
DEN	33	3.63	38	4.19
DET	22	2.42	44	4.85
GB	34	3.74	4	0.44
HOU	28	3.08		
IND	22	2.42	68	7.49
JAC	22	2.42	22	2.42
KC	30	3.30	61	6.72
LA	4	0.44		
MIA	21	2.31		
MIN	25	2.75	46	5.07
NE	27	2.97		
NO	29	3.19		
NYG	26	2.86	24	2.64
NYJ	23	2.53	23	2.53
OAK	25	2.75	62	6.83
PHI	29	3.19	32	3.52
PIT	35	3.85	17	1.87
SD	38	4.19		
SEA	34	3.74	61	6.72
SF	37	4.07	51	5.62
STL	29	3.19	56	6.17
TB	22	2.42	13	1.43
TEN	27	2.97		
WAS	27	2.97	48	5.29
Observations	908		908	

CRediT authorship contribution statement

Brad R. Humphreys: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Jane E. Ruseski:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Project administration.

Data availability

Data will be made available on request.

Appendix

A.1. Sample characteristics

The sample of games analyzed does not include all NFL games played (see Tables 9 and 11).

Table 11
PM2.5 data coverage by stadium/season (number of plays).

Stadium	2008	2009	2010	2011	2012	2013	2014	2015	2016	Total
Alameda County Coliseum	0	2,175	8,425	8,712	8,952	16,821	25,166	25,450	29,080	124,781
Arrowhead Stadium	0	1,974	7,946	8,673	8,497	12,104	17,064	17,246	16,617	90,121
CenturyLink Field	0	0	6,538	16,658	17,029	16,831	18,461	25,453	25,166	126,136
Cleveland Browns Stadium	0	0	0	0	6,513	8,653	5,810	3,716	8,705	33,397
Edward Jones Dome	0	4,359	8,360	21,624	32,093	41,871	40,417	41,141	42,463	232,328
EverBank Field	0	0	0	0	0	0	4,900	5,577	7,204	17,681
FedEx Field	0	0	0	7,353	8,400	8,427	8,597	13,761	14,012	60,550
Ford Field	0	0	7,931	8,518	8,245	7,831	7,351	7,579	0	47,455
Georgia Dome	0	0	0	7,856	7,608	7,340	8,195	8,429	6,614	46,042
Heinz Field	0	0	8,529	3,449	0	0	0	0	15,048	27,026
Lambeau Field	0	0	4,562	4,694	0	0	0	0	0	9,256
Levi's Stadium	0	2,194	7,046	0	2,184	8,598	11,400	17,098	17,153	65,673
Lincoln Financial Field	0	0	0	0	0	16,895	31,050	30,540	40,476	118,961
Lucas Oil Stadium	0	2,185	15,541	20,252	14,929	14,109	13,552	16,289	25,817	122,674
M&T Bank Stadium	4036	8,569	8,513	8,552	8,549	8,302	8,399	7,987	8,423	71,330
Metrodome	0	0	0	4,126	8,168	19,254	34,272	15,442	25,740	107,002
Meadowlands Stadium	0	0	0	0	0	0	7,089	17,097	32,314	56,500
Paul Brown Stadium	0	0	3,500	8,570	4,036	0	0	6,454	8,682	31,242
Raymond James Stadium	0	0	0	0	0	0	0	8,253	6,067	14,320
Sports Authority Field at Mile High	0	0	0	0	0	6,462	16,968	25,607	37,753	86,790
University of Phoenix Stadium	0	0	5,572	21,724	23,018	24,529	25,905	28,514	27,419	156,681
Total	4036	21,456	92,463	150,761	158,221	218,027	284,596	321,633	394,753	1,645,946

PM 2.5 data 2008–2016. Data from closest air quality monitoring station within 10 miles of stadium.

2008 has only games played at M&T Bank Stadium in Baltimore.

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